

Deep Learning

Theoretical Exercises - Lesson 1 - Chapter 2

Exercises on the book "Deep Learning" written by Ian Goodfellow,
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Exercises and solutions by T. Méndez and G. Schuster

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1 Exercises on Linear Algebra

1. Calculate the following matrix products.

(a) $C = AB$ and $D = BA$

$$\text{with } A = \begin{pmatrix} 7 & 7 & 0 \\ 8 & 4 & 2 \end{pmatrix} \text{ and } B = \begin{pmatrix} 7 & 0 \\ 0 & 1 \\ 3 & 8 \end{pmatrix}$$

(b) $b = Ax$

$$\text{with } A = \begin{pmatrix} 7 & 7 & 0 \\ 8 & 4 & 2 \end{pmatrix} \text{ and } x = \begin{pmatrix} 7 \\ 3 \\ 10 \end{pmatrix}$$

(c) $c = h^T A$

$$\text{with } A = \begin{pmatrix} 7 & 7 & 0 \\ 8 & 4 & 2 \end{pmatrix} \text{ and } h = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

2. Calculate the inverse of the orthogonal matrix

$$A = \frac{1}{3} \begin{pmatrix} 2 & -2 & 1 \\ 1 & 2 & 2 \\ 2 & 1 & -2 \end{pmatrix}.$$

and check whether your result is correct. Also find the solution of $Ax = b$ with

$$b = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}$$

using the inverse matrix.

3. Calculate the L^p norm of the vector $y = (4 \quad -3)^T$ for $p = 1, 2, 3$.

4. The following overdetermined, linear system of equations $A\mathbf{x} = \mathbf{b}$ is given, where

$$A = \begin{pmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 6 \end{pmatrix}, \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} 1.8 \\ 3.3 \\ 4.1 \end{pmatrix}.$$

Since there are more linear independent equations than unknowns, there is no solution. For this reason, the most reasonable solution should be found, by using the Moore-Penrose pseudoinverse

$$A^+ = VD^+U^T,$$

for which a singular value decomposition must be carried out. To perform a singular value decomposition, follow the steps below.

- (a) Determine the eigenvalues λ_1 and λ_2 of $\mathbf{B} = A^T A$, where $\lambda_1 > \lambda_2$.
- (b) Determine the Matrix D as $D = \begin{pmatrix} \sqrt{\lambda_1} & 0 \\ 0 & \sqrt{\lambda_2} \\ 0 & 0 \end{pmatrix}$
- (c) Determine matrix $V = (\mathbf{v}^{(1)} \quad \mathbf{v}^{(2)})$, where $\mathbf{v}^{(i)}$ are the eigenvectors of $\mathbf{B} = A^T A$. Note that $\mathbf{v}^{(1)}$ is the eigenvector to the larger eigenvalue λ_1 .
- (d) Determine matrix $U = (\mathbf{u}^{(1)} \quad \mathbf{u}^{(2)} \quad \mathbf{u}^{(3)})$, where $\mathbf{u}^{(i)}$ are the eigenvectors of $\mathbf{C} = AA^T$. Note that $\mathbf{u}^{(1)}$ is the eigenvector to the largest eigenvalue and $\mathbf{u}^{(3)}$ is the eigenvector to the smallest eigenvalue.
- (e) Check whether U and V are orthogonal matrices and whether $A = UDV^T$
- (f) Determine the Matrix D^+ .
- (g) Determine the Moore-Penrose pseudoinverse A^+ .
- (h) Determine the most reasonable solution $\mathbf{x}$.



Hint:

Use the calculator or python to determine the eigenvalues and eigenvectors.